Public electric-vehicle charging and accessibility inequality in Bogotá: A socio-spatial allocation framework

Abstract

Public electric-vehicle charging is both a decarbonization challenge and a transport- geography problem because infrastructure location redistributes accessibility across urban space. This study examines charger allocation in Bogotá through a framework that links spatio-temporal graph learning and accessibility-based lexicographic optimization. Official registration checkpoints show that the city's EV fleet grew from 2,039 vehicles in November 2020 to 8,899 in November 2024, increasing pressure to allocate public charging capacity across 142 UTAM zones. Household travel survey data, socioeconomic attributes, and network travel times are encoded as an urban graph and processed through a spatio-temporal graph neural network (ST-GNN) to estimate zonal priorities and venue-specific demand structure under fixed quotas. We then test whether charger deployment can remain consistent with mobility-derived need while improving access in socially vulnerable areas. Relative to territorial equalization, the socially weighted scenario changes allocations in 10 zones, raises coverage of vulnerable UTAMs within 15 minutes from 86.11% to 88.89%, and reduces the vulnerability-weighted accessibility Gini from 0.197041 to 0.196896. Relative to the priority-matching baseline, that improvement entails an 8.0% increase in priority- deviation loss. The soft-floor extension serves as a diagnostic robustness check rather than an additional redistributive scenario: it does not change the reproduced allocation, and 95.45% of protected zones already satisfy the calibrated minimum-service floor. The results show how transport-geography concepts can be operationalized in charging planning without disconnecting decisions from observed mobility structure.

Keywords: electric-vehicle charging, transport geography, accessibility, socio-spatial inequality, urban mobility, Bogotá, graph neural networks

1. Introduction

Electric-vehicle (EV) charging infrastructure is central to urban transport transitions (Hopkins et al., 2023; Varghese et al., 2024), but it is also a transport geography question: where infrastructure is placed, how access varies, and who benefits. This matters most where renters, apartment dwellers, and households without off-street parking depend on public charging (Hardman et al., 2018; Neaimeh et al., 2017; Soltani Mandolakani and Singleton, 2024; Chen et al., 2025). In such settings, charger allocation can reinforce or mitigate socio-spatial inequality (Carlton and Sultana, 2024; Yu et al., 2025). Recent evidence further shows that charger-access disparities can be especially acute in neighborhoods with more multi-unit dwellings and among households that must reorganize daily travel around limited charging options (Hsu and Fingerman, 2021; Gan, 2023; Malabanan et al., 2026).

The challenge is sharper in data-constrained cities, where planners must rely on household travel surveys, mobility indicators, and socioeconomic data rather than detailed charging telemetry. Related work in this journal shows that accessibility indicators, Lorenz-curve diagnostics, and transport investment appraisal can reveal sharply different equity patterns across urban contexts (Geurs and van Wee, 2004; Delbosc and Currie, 2011; Bocarejo and Oviedo, 2012). Existing work also points to three relevant strands, but these are often treated separately: equity-oriented accessibility analysis (Hopkins et al., 2023; Varghese et al., 2024; Jiao et al., 2024; Lezcano et al., 2025), spatio-temporal charging-demand learning (Wang et al., 2023a; Çavuş et al., 2025; Wang et al., 2023b, 2025), and optimization-based charger allocation (Zhang et al., 2023; Qiao et al., 2023; Zhou et al., 2023; Li et al., 2024; Sun et al., 2025). Recent review work further synthesizes the forecasting literature (Alaraj et al., 2025).

Recent contributions in the journal further show that accessibility interpretation is sensitive to zonal aggregation (Qu et al., 2026) and that EV travel behavior varies with the built environment in nonlinear ways (Tian et al., 2026), reinforcing the need for geographically differentiated planning.

This paper addresses that gap by treating public charging as a socio-spatial allocation problem. It links the stages of one spatial decision chain: mobility and socioeconomic data are encoded as an urban graph; an ST-GNN extracts zone-level priority signals and temporal attention patterns; those outputs are translated into optimization-ready targets and venue- specific shares; and the resulting allocations are evaluated through accessibility and equity indicators.

The empirical motivation is Bogotá, where charging ports must be allocated across urban zones under fixed quotas, partial behavioral observability, and uneven social disadvantage (Secretaría Distrital de Movilidad, 2025). In this context, the framework distinguishes territorial accessibility equalization from socially weighted equalization and tests whether a soft minimum-service rule becomes binding once vulnerability-sensitive allocation is introduced.

Accordingly, the analysis asks three linked questions: how spatio-temporal graph learning reshapes zone-level charging priorities; what trade-offs emerge between learned demand fidelity and socio-spatial equity; and how territorial, socially weighted, and soft minimum-service rules alter outcomes in vulnerable zones.

The paper makes three contributions. First, it converts graph-learned mobility signals into auditable optimization targets. Second, it quantifies the trade-off between demand fidelity and socio-spatial accessibility improvement under fixed quotas, emphasizing that even modest aggregate shifts can matter when allocations are integer-valued and spatially constrained. Third, it offers a practical template for data-constrained cities seeking to align charging deployment with both mobility structure and social need. The article's objective is not to claim state-of-the-art forecasting performance in its own right, but to show how learned mobility signals can be translated into transparent, policy-auditable allocation rules.

2. Literature Review

2.1. Electric-vehicle charging as a transport geography question

EV charging equity analysis has progressively moved from charger-count metrics toward accessibility and justice-oriented indicators (Hopkins et al., 2023; Varghese et al., 2024). In transport geography, foundational work on accessibility and equity provides the interpretive basis for that shift (Geurs and van Wee, 2004; Delbosc and Currie, 2011; Bocarejo and Oviedo, 2012). This shift is crucial because equal numbers of chargers across zones do not imply equal effective access once travel time, service type, and user constraints are considered (Jiao et al., 2024; Lezcano et al., 2025; Peng et al., 2024). Recent empirical work further shows that charger access gaps can be sharper across race, income, and housing tenure, especially in areas with more multi-unit housing and limited home-charging options (Hsu and Fingerman, 2021; Gan, 2023; Malabanan et al., 2026). Recent evidence also shows that spatially balanced infrastructure can remain socially uneven when vulnerable households face structural constraints on private charging (Carlton and Sultana, 2024; Yu et al., 2025; Chen et al., 2025). These studies show that charging provision should be evaluated through the geography of access rather than charger counts alone, and that zonal accessibility measures may still miss socially meaningful disparities when vulnerability and housing constraints vary across urban space. Recent work in this journal extends that agenda in several directions: combined perceived and modeled charger accessibility in London (Zhang and Cao, 2025), multi-scalar regional inequity in the Greater Bay Area (Gao et al., 2025), national disparities across the United States (Bashar et al., 2026), and workplace-sensitive accessibility assessment (Cai et al., 2026). Recent work also shows that aggregation choices can distort accessibility diagnosis and equity interpretation (Qu et al., 2026).

2.2. Learning spatially differentiated mobility demand

Graph-based and spatio-temporal learning models are increasingly used to capture urban mobility dependence structures that simpler models cannot represent (Wang et al., 2023a; Çavuş et al., 2025). These approaches are well suited for charging planning because mobility demand is both networked in space and dynamic in time. Complementary work on forecasting and integrated planning further supports the value of learning-based demand signals for infrastructure decisions (Wang et al., 2023b; Alaraj et al., 2025; Wang et al., 2025). Evidence on mixed charging-behavior patterns also reinforces the relevance of temporal heterogeneity in charging demand (Lee et al., 2020). Recent evidence from the journal also indicates that EV adoption and use respond nonlinearly to built-environment conditions (Tian et al., 2026). In data-constrained contexts, however, the key issue is not only predictive performance but how learned signals are converted into optimization-ready targets.

2.3. Locational optimization, territory, and accessibility equity

Optimization-based charger planning has evolved from static siting toward richer formulations that incorporate user behavior, network effects, and fairness constraints (Zhang et al., 2023; Qiao et al., 2023; Zhou et al., 2023; Li et al., 2024; Sun et al., 2025). Lexicographic and hierarchical designs are particularly relevant when planners must preserve operational priorities while improving equity outcomes.

At the same time, transport-equity research emphasizes that social disadvantage is multi-dimensional and interacts with spatial accessibility (Lucas et al., 2016; Delmelle and Casas, 2012). Consequently, optimization models that only equalize zonal outcomes may underperform in socially heterogeneous cities. This supports introducing vulnerability-sensitive objectives and service floors.

2.4. Research gap

The literature therefore points to a missing bridge: an explicit framework that connects learning outputs, decision rules, and distributive outcomes within one auditable planning chain. This matters especially in data-constrained cities, where planners must justify how indirect mobility evidence is translated into allocation targets and accessibility outcomes.

3. Study area, data and methods

This section defines the data, learning model, target extraction, and allocation scenarios.

3.1. Study area and data

The case study is Bogotá, represented by 142 UTAM zones, consistent with local planning instruments (Secretaría Distrital de Movilidad, 2025). The objective is to allocate venue-specific charging ports across zones while preserving interpretable links to mobility demand and accessibility equity.

Official registration checkpoints situate this allocation problem within a rapidly expanding EV transition. The citywide

stock increased from 2,039 registered EVs in November 2020 to 8,899 in November 2024, with a mean annual addition of 1,715 vehicles and the largest observed increment (+2,479) between October 2023 and November 2024. Figure 1 summarizes this empirical trajectory.

The analysis combines four data blocks: household travel survey records and mobility profiles, EV planning baselines and venue quotas (Wood, 2022), socioeconomic and housing proxies for charging vulnerability, and road-network travel times. Table 1 summarizes each source and role.

Because zone-level charger transactions are not observed in this setting, the learning target is a survey-based proxy demand intensity rather than realized charging telemetry. The learning stage is therefore used to recover relative spatio-temporal priorities under partial observability, not to claim telemetry-equivalent prediction of individual charging events. Feature construction, preprocessing, train–validation partitioning, and implementation settings used for reproduction are documented in the blinded replication package. The validation diagnostics reported below are computed on held-out time steps outside T_{train}, so the learning evaluation does not reuse the observations used to estimate the network parameters. The main text can therefore remain focused on the learning-to-decision chain.

3.2. Problem framing and methodological overview

The planning problem is to assign integer charging ports n_{iv} to each zone $i \in I$ and venue type $v \in V$ under fixed venue quotas Q_v, while balancing two objectives: (i) fidelity to mobility-derived priorities and (ii) equity in effective accessibility across urban space.

Figure 2 summarizes the staged workflow from data assembly to policy interpretation.

This framing keeps the main policy parameters (ε, κ, γ, β) explicit. The next subsections formalize that chain.

3.3. Graph representation and feature construction

To encode urban interaction structure, we represent the city as a weighted graph $\mathcal{G} = (I, E, \mathbf{W})$, where nodes are zones and edges represent feasible mobility or accessibility interactions.

The weighted adjacency matrix is defined as:

$$w_{ij} = \exp(-t_{ij}/\tau), \tag{1}$$

where t_{ij} is network travel time from zone i to zone j, and τ is a scale parameter controlling distance decay. In the final implementation, this graph-based decay was used only in the learning stage, whereas the optimization stage employed venue-specific decay parameters λ_v in the accessibility function.

For each zone and time step, the node-feature vector is:

$$\mathbf{x}_{i,t} = \left[d_{i,t}^{(1)}, \ldots, d_{i,t}^{(P)}, q_i^{(1)}, \ldots, q_i^{(R)}\right]^\top, \tag{2}$$

where $d_{i,t}^{(p)}$ are time-varying mobility descriptors and $q_i^{(r)}$ are static zone attributes (including socioeconomic and built-environment factors). Here P and R denote the numbers of time-varying and static features, respectively. Stacking all zones gives $\mathbf{X}_t \in \mathbb{R}^{|I|\times(P+R)}$.

To keep the main text focused on model logic, notation used in Sections 3.3–3.5 is summarized in Supplementary Appendix A.

This representation is required for the next step, where spatial propagation and temporal encoding jointly extract decision-relevant mobility signals.

3.4. Spatio-temporal GNN model

Given $\mathcal{G}$ and $\mathbf{X}_t$, spatial dependence is modeled through message passing:

$$\mathbf{h}_{i,t}^{(l+1)} = \sigma\left(\mathbf{W}_s^{(l)}\mathbf{h}_{i,t}^{(l)} + \sum_{j\in\mathcal{N}(i)} w_{ij}\mathbf{U}_s^{(l)}\mathbf{h}_{j,t}^{(l)}\right), \tag{3}$$

where $\mathcal{N}(i)$ denotes the neighbors of zone i, $l = 1, \ldots, L$ indexes the spatial message-passing layers, $\mathbf{W}_s^{(l)}$ and $\mathbf{U}_s^{(l)}$ are trainable weight matrices, and $\sigma(\cdot)$ is a nonlinear activation.

Temporal dynamics are then captured by a recurrent update:

$$\mathbf{g}_{i,t} = \text{GRU}\left(\mathbf{h}_{i,t}^{(L)}, \mathbf{g}_{i,t-1}\right), \tag{4}$$

and the demand signal is read out as:

$$\hat{y}_{i,t} = f_{\text{out}}(\mathbf{g}_{i,t}). \tag{5}$$

where $f_{\text{out}}(\cdot)$ is the output head of the network.

To identify behaviorally relevant periods, we compute temporal attention weights:

$$\alpha_t = \frac{\exp\left(\mathbf{q}^\top \tanh(\mathbf{W}_a\mathbf{r}_t)\right)}{\sum_{\tau\in T}\exp\left(\mathbf{q}^\top \tanh(\mathbf{W}_a\mathbf{r}_\tau)\right)}, \qquad \mathbf{r}_t = \frac{1}{|I|}\sum_{i\in I}\mathbf{g}_{i,t}, \tag{6}$$

where $\mathbf{r}_t$ is the mean latent representation at time t, and $\mathbf{q}$ and $\mathbf{W}_a$ are trainable attention parameters. We then aggregate zone-level raw priority as:

$$s_i^{\text{raw}} = \sum_{t\in T}\alpha_t\hat{y}_{i,t}. \tag{7}$$

Model parameters are estimated by minimizing:

$$\mathcal{L}_{\text{GNN}} = \frac{1}{|I||T_{\text{train}}|}\sum_{t\in T_{\text{train}}}\sum_{i\in I}(\hat{y}_{i,t} - y_{i,t})^2. \tag{8}$$

where $T_{\text{train}} \subseteq T$ is the subset of time steps used for training.

In this paper, the learning model is treated as a priority-extraction component within a broader planning chain. Predictive diagnostics are reported for transparency, but the main substantive claim of the article concerns the auditable translation from learned signals to allocation rules rather than algorithmic dominance over simpler forecasting baselines.

Equation (3) captures spatial spillovers. Equation (4) captures temporal dependence. Equations (6)–(7) convert predictions into interpretable priority signals. This output is the direct input to the allocation model in the next subsection.

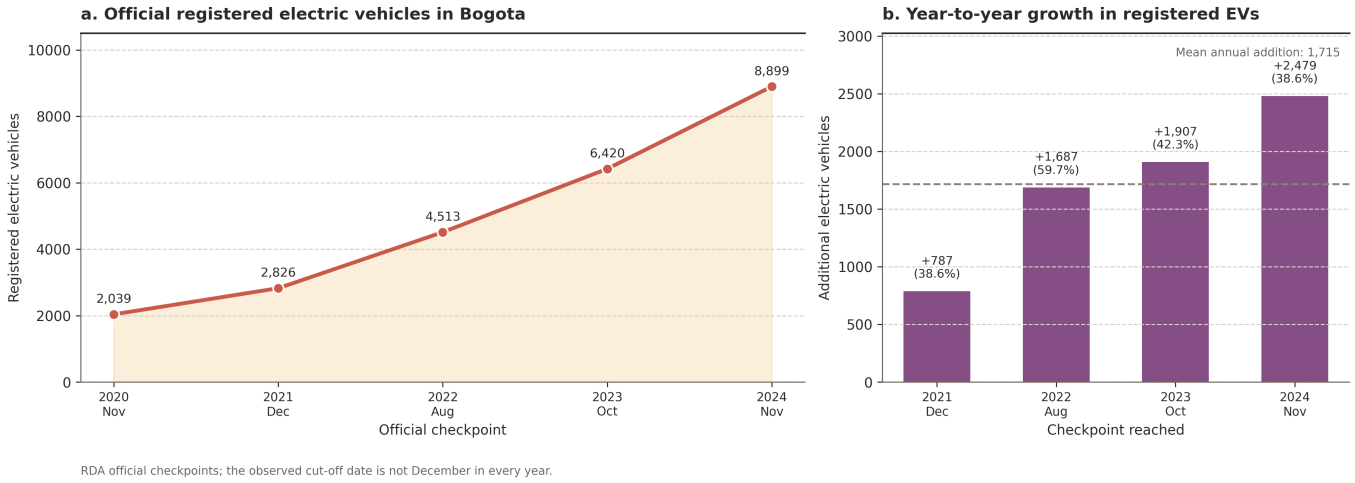


Figure 1: Official registration checkpoints for electric vehicles in Bogotá and year-to-year additions. Checkpoint dates do not always coincide with December year-end.

Table 1: Core datasets and analytical roles.

Source	Role in model chain	Spatial unit	Key variables
Household travel survey	Mobility features for graph learning and behavioral interpretation	Zone/time	Origin–destination flows, departure times, trip purposes
Socioeconomic and housing records	Vulnerability index for social-equity modeling	Zone	Income, rental tenure, apartment share, parking proxy
Road network or OD time matrix	Accessibility impedance and graph weights	Zone-to-zone	Travel times, generalized impedance
Existing EV planning framework	Exogenous quotas and benchmarking signals	City/zone	Venue quotas, baseline demand-priority references

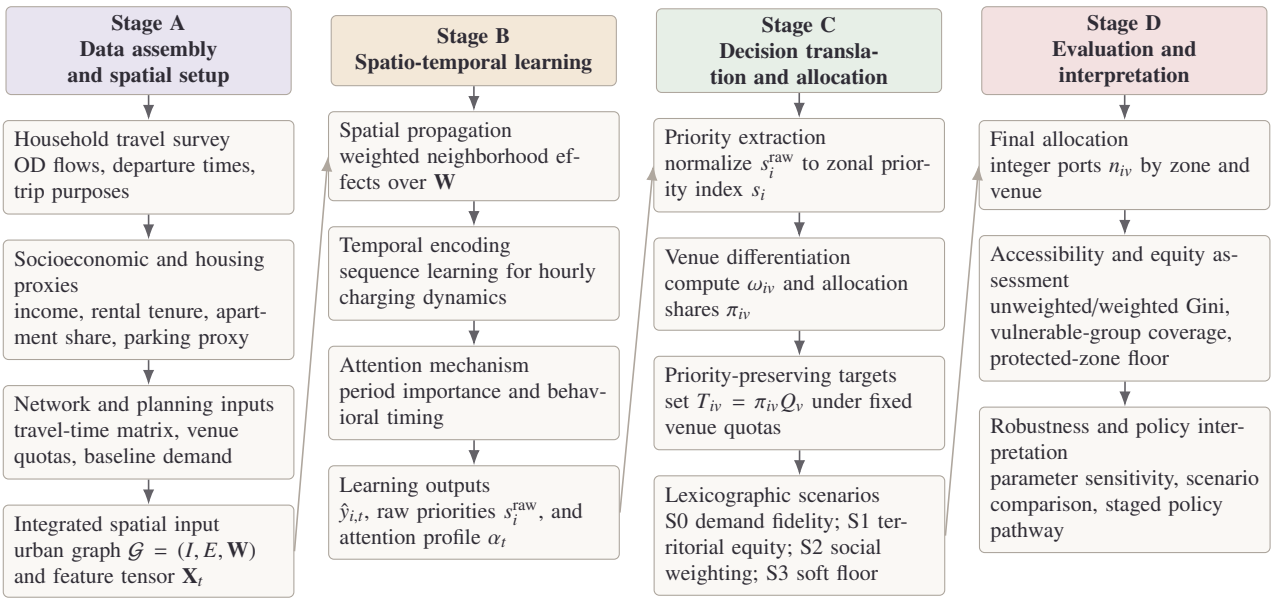


Figure 2: Staged methodology from data assembly to accessibility outcomes. Stages A–D distinguish input assembly, spatio-temporal learning, translation into allocation scenarios, and evaluation/policy interpretation.

3.5. Priority extraction and behavioral interpretation

The optimization model requires normalized and venue-specific signals rather than raw predictions. We therefore normalize raw priorities into [0, 1]:

$$s_i = \frac{s_i^{\text{raw}} - \min_{k\in I} s_k^{\text{raw}}}{\max_{k\in I} s_k^{\text{raw}} - \min_{k\in I} s_k^{\text{raw}}}. \tag{9}$$

If the raw-priority range were to collapse to zero, the implementation would revert to a uniform zonal priority vector; this safeguard was not triggered in the reproduced run.

Venue-specific weights are extracted by aggregating attention-weighted predictions over venue-relevant periods T_v:

$$\omega_{iv} = \frac{\sum_{t\in T_v} \alpha_t \hat{y}_{i,t}}{\sum_{v'\in V} \sum_{t\in T_{v'}} \alpha_t \hat{y}_{i,t}}. \tag{10}$$

Here $T_v \subseteq T$ denotes the venue-relevant time window for venue type v; in the empirical implementation these windows are mutually exclusive and correspond to residential, work, and public-service periods. If the denominator in Eq. (10) were zero for a zone, the implementation would assign uniform venue weights; this safeguard was not triggered in the reproduced run.

These quantities define optimization shares:

$$\pi_{iv} = \frac{s_i \omega_{iv}}{\sum_{k\in I} s_k \omega_{kv}}, \qquad T_{iv} = \pi_{iv} Q_v. \tag{11}$$

Equation (9) preserves relative ranking across zones. Equation (10) maps temporal behavior into venue differentiation. Equation (11) translates learning outputs into target allocations. This is the explicit GNN → optimization bridge used in Section 3.6.

3.6. Two-stage lexicographic optimization model

The allocation model is solved lexicographically. First, it minimizes deviation from priority-preserving targets T_{iv}. Second, it minimizes accessibility inequality while allowing only bounded deterioration in priority fidelity.

Optimization notation is summarized in Supplementary Appendix B. In scenarios S1–S3, the lower-level problem is solved over the same feasible set defined by Eqs. (15)–(18); the lower-level constraints introduced below are added to those upper-level requirements rather than replacing them.

Social vulnerability is defined as:

$$\nu_i = \frac{\sum_{m\in M} \eta_m z_{im} - \min_{k\in I} \sum_{m\in M} \eta_m z_{km}}{\max_{k\in I} \sum_{m\in M} \eta_m z_{km} - \min_{k\in I} \sum_{m\in M} \eta_m z_{km}}. \tag{12}$$

Accessibility is computed as:

$$A_i = \sum_{v\in V} \omega_{iv} \sum_{j\in I} \exp(-t_{ij}/\lambda_v) n_{jv}. \tag{13}$$

Upper level (priority fidelity)..

$$\min U = \sum_{i\in I} \sum_{v\in V} u_{iv} \tag{14}$$

subject to

$$u_{iv} \geq n_{iv} - T_{iv} \qquad \forall i \in I, \forall v \in V, \tag{15}$$

$$u_{iv} \geq T_{iv} - n_{iv} \qquad \forall i \in I, \forall v \in V, \tag{16}$$

$$\sum_{i\in I} n_{iv} = Q_v \qquad \forall v \in V, \tag{17}$$

$$n_{iv} \in \mathbb{Z}_+,\ u_{iv} \geq 0 \qquad \forall i \in I, \forall v \in V. \tag{18}$$

The objective function minimizes total deviation from GNN-derived target allocations. Constraint (15) and Constraint (16) linearize absolute deviations. Constraint (17) enforces venue-specific quota conservation. Constraint (18) defines variable domains.

Lower level (equity objective with bounded fidelity loss).. Let $U^\star$ be the optimum of Eq. (14). Accessibility-gap linearization is:

$$z_{ij} \geq A_i - A_j \qquad \forall i < j, \tag{19}$$

$$z_{ij} \geq A_j - A_i \qquad \forall i < j, \tag{20}$$

$$z_{ij} \geq 0 \qquad \forall i < j. \tag{21}$$

The unweighted and socially weighted equity objectives are, respectively:

$$\min E_{\text{U}} = \sum_{i<j} z_{ij}, \tag{22}$$

$$\min E_{\text{W}} = \sum_{i<j} \bar{\nu}_{ij} z_{ij}, \qquad \bar{\nu}_{ij} = 1 + \kappa \frac{\nu_i + \nu_j}{2}. \tag{23}$$

Near-optimal coupling is imposed as:

$$\sum_{i\in I} \sum_{v\in V} u_{iv} \leq (1+\varepsilon) U^\star. \tag{24}$$

For the minimum-service scenario, the protected set $\mathcal{V}^H$ is defined as the top 15% of zones in the vulnerability distribution, operationalized in the reproduced run as the 22 highest-ranked zones under ν_i ($\lceil 0.15|I| \rceil$ with $|I| = 142$), and the accessibility floor is implemented as a soft constraint:

$$A_i + d_i \geq \underline{A} \qquad \forall i \in \mathcal{V}^H, \tag{25}$$

$$d_i \geq 0 \qquad \forall i \in \mathcal{V}^H. \tag{26}$$

In the empirical implementation, the floor is calibrated from the lower tail of the socially weighted baseline solution:

$$\underline{A} = \gamma\, q_{0.10}\Big(\{A_i^{S2} : i \in \mathcal{V}^H\}\Big), \tag{27}$$

where, after sorting $\{A_i^{S2} : i \in \mathcal{V}^H\}$ in ascending order, $q_{0.10}(\cdot)$ denotes the empirical order-statistic percentile $A^{S2,H}_{(\lceil 0.10|\mathcal{V}^H|\rceil)}$ on that discrete protected-set sample, and $0 < \gamma \leq 1$ is a floor-relaxation factor.

In scenario S3, the soft-floor shortfalls are penalized jointly with weighted accessibility inequality. Let $\beta > 0$ denote the shortfall penalty. The S3 objective is:

$$\min E_{\text{F}} = \sum_{i<j} \bar{\nu}_{ij} z_{ij} + \beta \sum_{i\in\mathcal{V}^H} d_i. \tag{28}$$

This formulation preserves the socially weighted structure of S2 while introducing an explicit lower-tail service guarantee.

For reporting, the inequality and coverage metrics are computed ex post from the realized accessibility vector A_i rather than read directly from the lower-level objective value. Let $\mathcal{V}^Q \subset I$ denote the most vulnerable quartile of zones (the top 25% of the ν_i distribution), operationalized in the reproduced run as the 36 highest-ranked zones under ν_i ($\lceil 0.25|I| \rceil$ with $|I| = 142$). The reported vulnerable-quartile share and vulnerable-zone coverage are:

$$S^Q = \frac{\sum_{i \in \mathcal{V}^Q} A_i}{\sum_{i \in I} A_i}, \tag{29}$$

$$C^{15} = \frac{1}{|\mathcal{V}^Q|} \sum_{i \in \mathcal{V}^Q} c_i^{15}, \tag{30}$$

$$c_i^{15} = \mathbf{1}\left[\exists j \in I : \; t_{ij} \le 15, \; \sum_{v \in V} n_{jv} > 0\right].$$

The unweighted accessibility Gini sorts $\{A_i\}_{i \in I}$ into ascending order $A_{(1)} \le \cdots \le A_{(|I|)}$, defines cumulative totals $C_k = \sum_{r=1}^{k} A_{(r)}$, and computes:

$$G = \frac{|I| + 1 - 2 \sum_{k=1}^{|I|} C_k / C_{|I|}}{|I|}. \tag{31}$$

The vulnerability-weighted accessibility Gini uses zone weights $1 + \nu_i$:

$$G_{\mathrm{W}} = \frac{\sum_{i \in I} \sum_{j \in I} (1 + \nu_i)(1 + \nu_j)|A_i - A_j|}{2\mu_\nu \left(\sum_{i \in I}(1 + \nu_i)\right)^2}, \tag{32}$$

$$\mu_\nu = \frac{\sum_{i \in I}(1 + \nu_i) A_i}{\sum_{i \in I}(1 + \nu_i)}.$$

Figure 7 uses the corresponding Lorenz diagnostics after sorting zones by A_i; in the weighted case, the horizontal axis is the cumulative share of $\sum_i (1 + \nu_i)$ and the vertical axis is the cumulative share of $\sum_i (1 + \nu_i) A_i$.

The objective functions minimize accessibility inequality in unweighted or vulnerability-weighted form. Constraint (24) links equity improvements to an explicit bound on priority-fidelity loss. Constraints (25) and (26) define a soft minimum-service guarantee for high-vulnerability zones, allowing limited shortfalls when a hard floor would make the model infeasible.

This lexicographic design operationalizes the full chain of the paper: learned priorities set the target, and optimization transforms that target into socially evaluated allocations.

3.7. Solution strategy and implementation

The model is implemented in Pyomo. Stage A trains the ST-GNN and exports $s_i, \omega_{iv}, \pi_{iv}, T_{iv}$. Stage B solves the two-stage lexicographic allocation model as a mixed-integer linear program on NEOS using CPLEX in the final experimental runs. The upper-level problem is solved first to obtain $U^\star$, after which lower-level equity scenarios are solved under the near-optimality constraint in Eq. (24).

Scenario definitions are shown in Table 2. In the final implementation, the accessibility kernel used $\lambda_{\mathrm{res}} = 15$, $\lambda_{\mathrm{work}} = 25$, and $\lambda_{\mathrm{pub}} = 10$ minutes. S1 used $\varepsilon = 0.03$, while S2 and S3 used $\varepsilon = 0.08$. The socially weighted objective used $\kappa = 4.0$. The protected set in S3 was defined as the top 15% of zones in the vulnerability distribution, implemented as the 22 highest-ranked zones under ν_i, and the soft floor was calibrated using $\gamma = 0.95$ and the empirical 10th percentile of S2 accessibility in that protected set. The shortfall penalty in S3 was set to $\beta = 100$. For reporting, the top-quartile set $\mathcal{V}^Q$ used in Eqs. (29)– (30) is distinct from the top-15% protected set $\mathcal{V}^H$ used only in S3 and is implemented as the 36 highest-ranked zones under ν_i. In the reproduced run, the S3 penalty term does not change the allocation relative to S2, even though one protected zone retains a positive shortfall. Accordingly, S3 is interpreted as a diagnostic soft-floor robustness check rather than as a substantively distinct redistribution outcome.

Performance is assessed with (i) priority-fidelity indicators, (ii) unweighted and weighted accessibility inequality, (iii) vulnerable-group coverage over the top vulnerability quartile within a 15-minute threshold, and (iv) robustness to decay, weighting, and policy assumptions. This metric design directly structures Section 4.

4. Results

The results are presented in the following order: empirical context, socio-spatial learning outputs, scenario allocations, distributional equity, and a compact policy trade-off interpretation.

4.1. Socio-spatial learning context and temporal attention

The ST-GNN produces a differentiated spatial priority surface rather than a uniform citywide demand map. Figure 3 shows that learned priority and social vulnerability only partially overlap: high-priority areas concentrate in a limited subset of the metropolitan footprint, whereas vulnerability is more diffusely distributed across the consolidated urban core. The protected geography used in S3 is likewise compact and localized, which makes targeted vulnerability-sensitive reallocation possible under fixed quotas.

Temporal structure is equally important. Figure 4 shows that the learned attention profile is strongest in the earliest hours of the day and then flattens through work and public-service windows. When disaggregated by social vulnerability, high-vulnerability zones receive relatively more weight in the first hours, whereas low-vulnerability zones retain a flatter daytime profile. Additional learning diagnostics report a best validation MAE of 10.333 and an attention-concentration index of 0.00257, indicating that attention remains diffuse rather than collapsing onto a single time slice. The MAE should be interpreted in the scale of the proxy-demand target rather than as a direct measure of observed charger transactions.

4.2. Scenario allocations and reallocation logic

Figure 5 shows that all four scenarios preserve the broad metropolitan structure of the learned allocation while redistributing ports within the consolidated urban core. Rank alignment between learned priority and installed ports remains high

Table 2: Policy scenarios used in the lexicographic allocation model.

Scenario	Policy role	ε	κ	Floor specification
S0	Priority-matching baseline	0.00	—	None
S1	Equity relaxation without vulnerability weights	0.03	0.0	None
S2	Socially weighted equity scenario	0.08	4.0	None
S3	Diagnostic soft-floor robustness check	0.08	4.0	Soft floor on top 15% protected set; $\gamma = 0.95$; $\beta = 100$

Notes: Relative fidelity losses versus S0 are 0.0%, 3.0%, 8.0%, and 8.0%, respectively. Coverage of vulnerable UTAMs within 15 minutes is C^{15} from Eq. (30), evaluated on the top vulnerability quartile $\mathcal{V}^{Q}$; this reporting set is distinct from the top-15% protected set $\mathcal{V}^{H}$ used only in S3. In the reproduced run, S3 reproduces the S2 allocation; the soft-floor penalty does not change n_{iv} relative to S2 and therefore acts as a diagnostic check rather than as an additional redistributive stage.

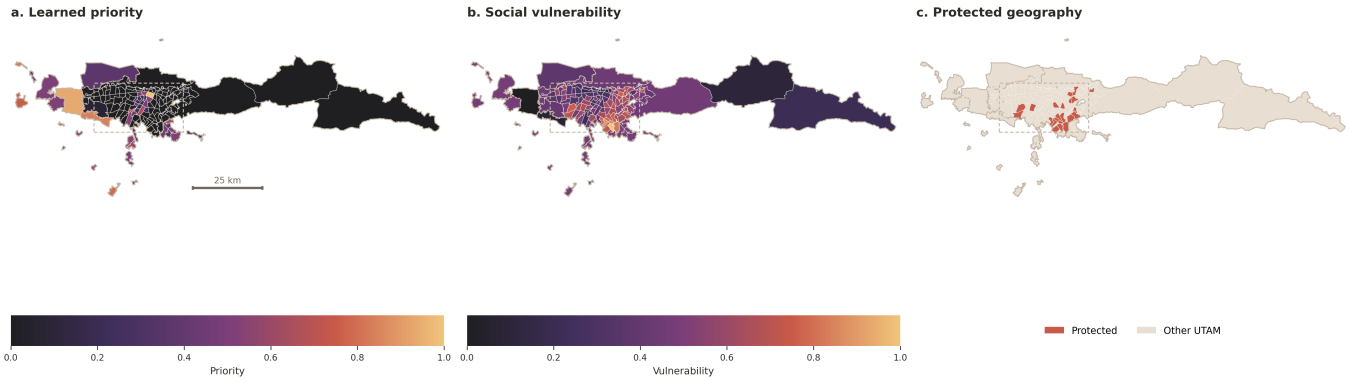


Figure 3: Socio-spatial inputs to the allocation stage: learned zonal priority, social vulnerability, and the protected geography used in the soft-floor scenario.

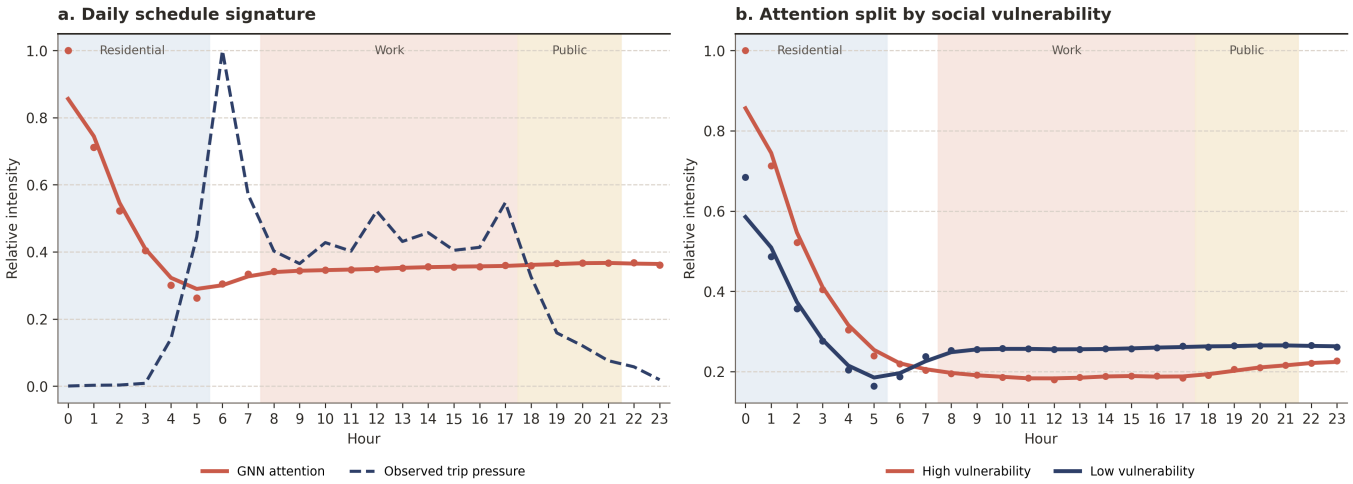


Figure 4: Temporal attention structure from the ST-GNN. Panel (a) compares hourly GNN attention with observed trip pressure across residential, work, and public windows; panel (b) splits the attention profile by social vulnerability.

across scenarios, with Spearman correlations of 0.922119 in S0, 0.916699 in S1, and 0.910472 in both S2 and S3. The absolute-deviation objective rises from 62.828438 in S0 to 64.713291 in S1 and 67.854713 in S2/S3, implying relative fidelity losses of 3.0% and 8.0%, respectively. Under fixed venue quotas and integer port assignments, these numerical changes are necessarily incremental, so even one-port reallocations should be interpreted as localized policy moves rather than as large system-wide redistributions.

The socially weighted scenario changes the allocation in 10 zones relative to S1. These are sparse, local reallocations—five one-port gains offset by five one-port losses—rather than a citywide reshuffling. Figure 6 makes this explicit. S3 is visually identical to S2 because the soft-floor term is non-binding in the reproduced run. The gains and losses form a compact cluster of spatially adjacent UTAMs in the consolidated inner urban area rather than a center–periphery inversion, which points to corridor-level rebalancing among neighboring zones with similar demand intensity but different vulnerability profiles.

4.3. Equity outcomes and policy trade-offs

Table 3 shows that socially weighted allocation improves the headline inequality and coverage indicators relative to both S0 and S1. Relative to territorial equalization (S1), the vulnerability-weighted accessibility Gini falls from 0.197041 to 0.196896, the unweighted Gini falls from 0.201248 to 0.201111, and vulnerable-zone coverage within 15 minutes rises from 86.11% to 88.89%. Mean accessibility in the protected high-vulnerability set changes only slightly (from 410.324 to 410.265), indicating that the equity gains come mainly from lower-tail rebalancing rather than large average increases. In relative terms, the reduction in the weighted Gini is small, which is consistent with the fact that the feasible policy move set is restricted to a limited number of integer reallocations under fixed quotas.

S3 does not produce additional allocational redistribution because it coincides with S2. The calibrated floor equals 287.140880 accessibility units, and 95.45% of protected zones already meet that threshold under the reproduced solution. The remaining total shortfall is 5.199 accessibility units, so the penalty term operates as a diagnostic safeguard without altering the final allocation in this run.

The slight decline in the share of accessibility captured by the most vulnerable quartile does not contradict the improvement in the weighted Gini. The quartile-share indicator is an aggregate concentration measure, whereas the lower-level objective directly targets pairwise dispersion and lower-tail shortfalls. In this reproduced run, the socially weighted scenario improves the lower tail mainly through rebalancing within and around vulnerable areas rather than by increasing the aggregate accessibility share captured by the vulnerable quartile as a whole.

Figure 7 shows the same pattern in standard transport-equity form. The equity-oriented scenarios shift both the unweighted and vulnerability-weighted Lorenz curves slightly toward equality, with the largest visual gain from S0 to S1 and a smaller additional gain under socially weighted allocation.

Table 4 and Figure 8 summarize the policy trade-off across scenarios. The move from S0 to S1 yields the largest marginal improvement in vulnerable coverage for a modest 3.0% fidelity loss, whereas the move from S1 to S2 adds a smaller but still positive improvement in coverage and the weighted Gini at higher fidelity cost. The robustness panels preserve the qualitative ordering around S2 across alternative decay, vulnerability-index, epsilon, kappa, and soft-floor specifications. S2 and S3 coincide because the calibrated soft-floor safeguard is non-binding in the reproduced allocation.

4.4. Sensitivity and policy robustness checks

Additional robustness checks indicate that the policy ordering is stable across plausible specification changes. Relative to the S2 baseline, the Spearman rank correlation of zone-level total allocated ports remains between 0.9958 and 1.0000 when the vulnerability index is reconstructed with PCA weights or the lexicographic tolerance varies between 0.056 and 0.100. The main sensitivity comes from the accessibility-decay parameters: relative to S2, reducing all λ_v values by 25% worsens the weighted Gini by 0.040144 and raises vulnerable coverage by 2.78 percentage points, whereas increasing them by 25% improves the weighted Gini by 0.018454 but lowers vulnerable coverage by 5.56 percentage points. Even in those cases, the same rank-stability measure remains high (0.9791–0.9839), so the qualitative scenario ordering is preserved. No alternative travel-time matrix with sufficient coverage was available for a full comparative rerun, so robustness should be read as conditional on the reproduced impedance structure.

5. Discussion

This study shows that charger planning becomes a transport-geography question once learned demand signals are evaluated through the spatial distribution of accessibility rather than through charger counts alone. The reproduced results reveal a clear but bounded trade-off. Priority-preserving allocation remains closest to the learned mobility surface, whereas vulnerability weighting redirects a small share of capacity toward zones where households are less able to rely on private charging. The important empirical result is therefore not a wholesale remapping of the city, but a selective redistribution within the consolidated urban core: only 10 zones change, yet vulnerable coverage rises from 86.11% to 88.89% and both weighted and unweighted accessibility inequality decline under fixed quotas. The aggregate inequality changes are numerically small, but this is exactly what should be expected when the feasible intervention is a limited set of one-port moves rather than a continuous reallocation of capacity.

That pattern matters for interpretation. It suggests that mobility-derived priority and social disadvantage overlap only partially, so equity gains come from rebalancing the lower tail of accessibility rather than abandoning the main demand geography. The contribution is thus substantive as well as methodological: it shows how a model trained on observed mobility can be translated into explicit allocation rules that remain

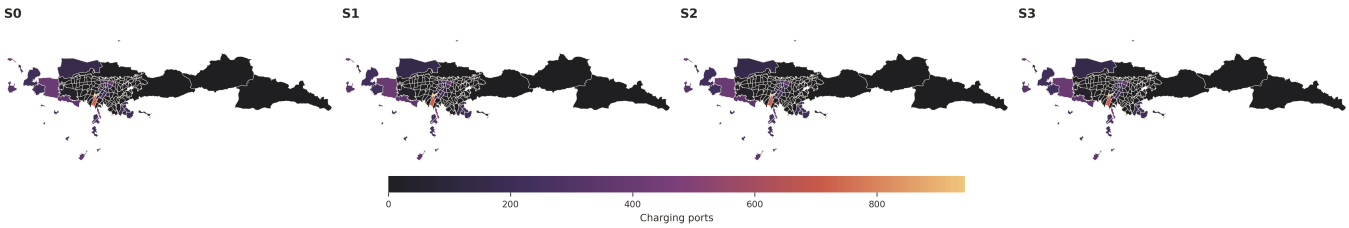


Figure 5: Scenario allocation maps for S0–S3. The soft-floor scenario S3 is visually identical to S2 because the floor term is non-binding in the reproduced solution.

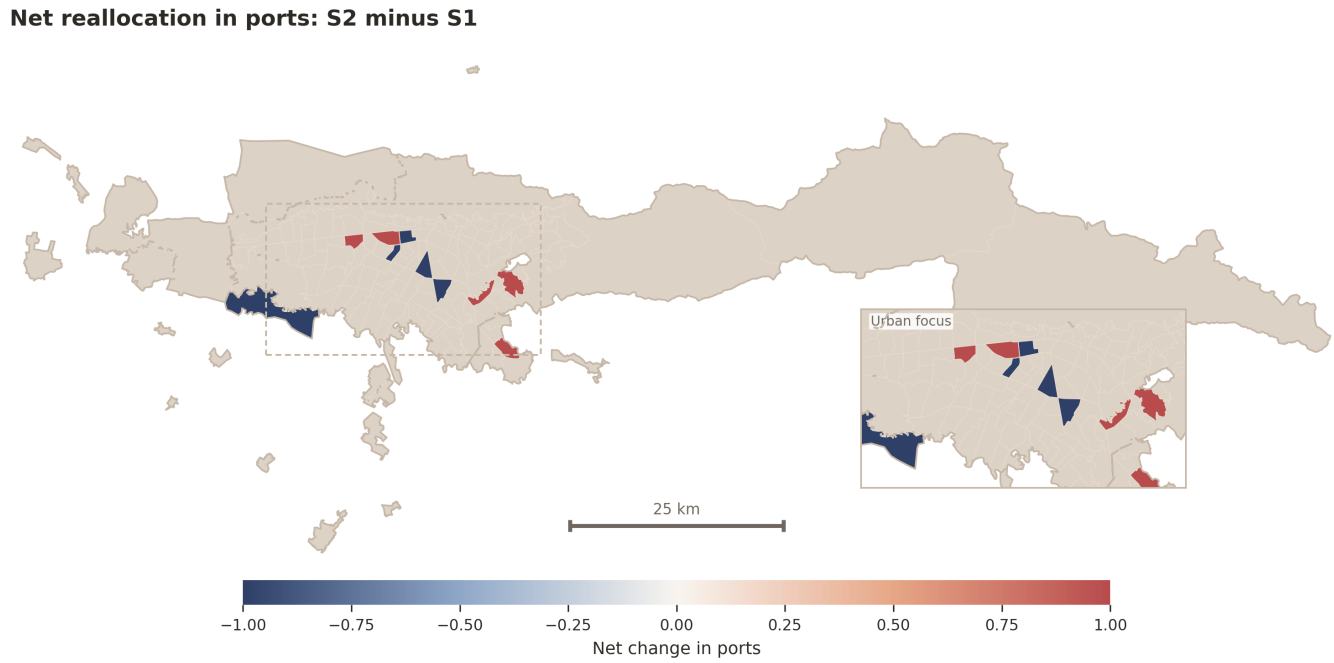


Figure 6: Net reallocation in charging ports under socially weighted equity relative to territorial equalization (S2 minus S1). Positive values indicate one-port gains and negative values indicate one-port losses.

Table 3: Accessibility and equity metrics by scenario.

Metric	S0	S1	S2	S3
Accessibility Gini (unweighted)	0.2015	0.2012	0.2011	0.2011
Accessibility Gini (vulnerability-weighted)	0.1972	0.1970	0.1969	0.1969
Share of accessibility captured by most vulnerable quartile	0.2725	0.2724	0.2723	0.2723
Coverage within 15 minutes for vulnerable UTAMs	0.7778	0.8611	0.8889	0.8889
Mean accessibility in protected set $\mathcal{V}^H$	410.68	410.32	410.26	410.26

Notes: In S3, the protected-set floor is calibrated at 287.14 accessibility units. The vulnerable-quartile share is S^{Q} from Eq. (29), coverage is C^{15} from Eq. (30), and the weighted Gini is G_{W} from Eq. (32). Although one protected zone retains a positive shortfall, the S3 allocation and aggregate equity indicators coincide with S2. The slight decline in the vulnerable-quartile share alongside lower Gini values reflects lower-tail rebalancing rather than greater aggregate concentration in the vulnerable quartile.

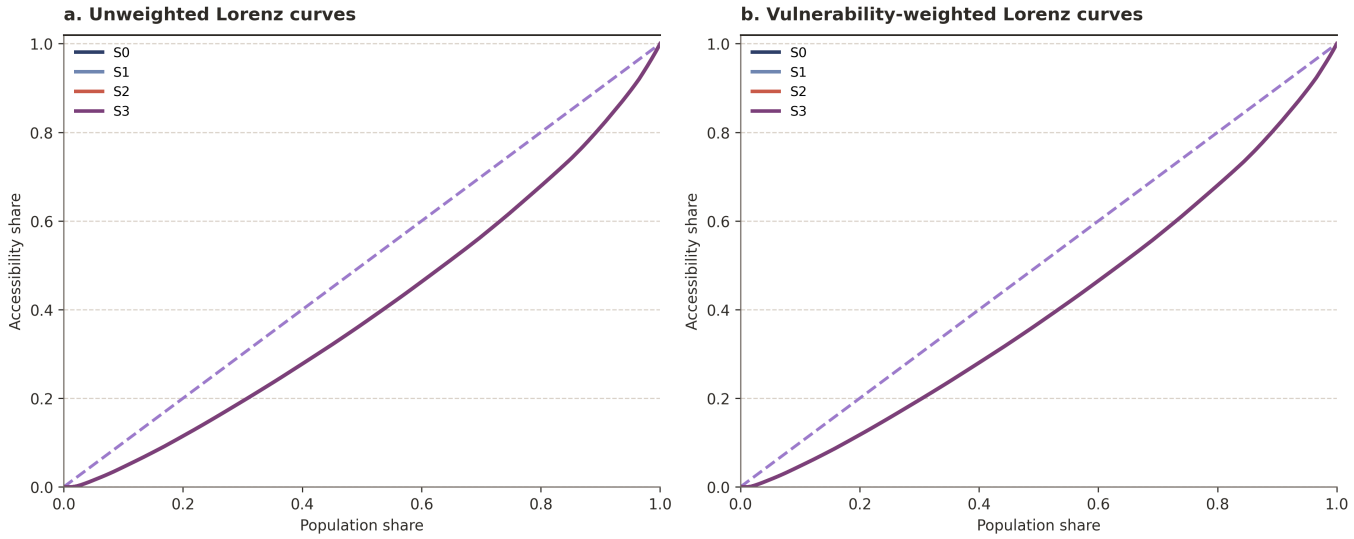


Figure 7: Unweighted and vulnerability-weighted accessibility Lorenz curves across scenarios.

Table 4: Compact policy trade-off summary across scenarios.

Scenario	Objective abs. deviation	Fidelity loss vs. S0 (%)	Vulnerable coverage (%)	Weighted Gini
S0	62.83	0.0	77.78	0.1972
S1	64.71	3.0	86.11	0.1970
S2	67.85	8.0	88.89	0.1969
S3	67.85	8.0	88.89	0.1969

Notes: S2 and S3 overlap exactly in the reproduced run. Relative to S1, the socially weighted scenario lowers the exact weighted Gini from 0.197041 to 0.196896 while changing mean accessibility in $\mathcal{V}^H$ from 410.324 to 410.265. The quantitative shift is modest in aggregate terms, but it is generated by only ten one-port reallocations under fixed quotas.

readable to planners, auditable in policy terms, and sensitive to the uneven geography of charging disadvantage. For data-constrained cities, the framework remains practical because it relies on surveys, socioeconomic records, and network travel times rather than charging telemetry.

5.1. Geographical implications and implementation pathway

The scenario sequence $S1 \rightarrow S2 \rightarrow S3$ can still be read as a staged governance pathway, but the reproduced run sharpens its interpretation. Territorial equalization comes first as the lowest-friction equity adjustment under fixed quotas. Vulnerability weighting follows once proxy evidence on charging disadvantage is available and politically defensible. Soft floors, in turn, operate as monitoring safeguards and auditing devices in the reproduced configuration; stronger redistributive use would require additional sensitivity analysis over β, γ, and the size of the protected set.

The minimum reproducible data package is modest: zonal mobility indicators, socioeconomic vulnerability proxies, an OD travel-time matrix, externally defined charger quotas, and a defensible citywide understanding of EV growth. In Bogotá, the official trajectory from 2,039 registered EVs in late 2020 to 8,899 in late 2024 underscores why these allocation decisions are no longer marginal, even in settings where detailed charging telemetry is not yet available.

The framework also has limitations. Vulnerability indicators are proxy-based and measured at the UTAM level, official EV-growth checkpoints are irregular rather than monthly, scenario outcomes depend on decay and weighting assumptions, and the current setup is static rather than multi-period. Transferability therefore depends less on reproducing the exact model architecture than on securing consistent zoning, a defensible travel-time matrix, minimal vulnerability proxies, externally defined quotas, and institutional capacity to interpret scenario-based optimization results. Because the article prioritizes the learning-to-decision translation over algorithmic competition, it also does not present an exhaustive benchmark contest against simpler forecasting models; the substantive claims should therefore be read as conditional on the reproduced learning stage rather than as a claim of universal model superiority.

Overall, the central implication is both methodological and geographical: learning-derived demand signals become policy-ready only when their translation into optimization parameters is explicit, auditable, and socially interpretable.

6. Conclusions

This paper shows that EV charger planning can move from territorial equalization toward social equity through a unified learning-and-optimization approach. In the reproduced run, the socially weighted scenario changes allocations in 10

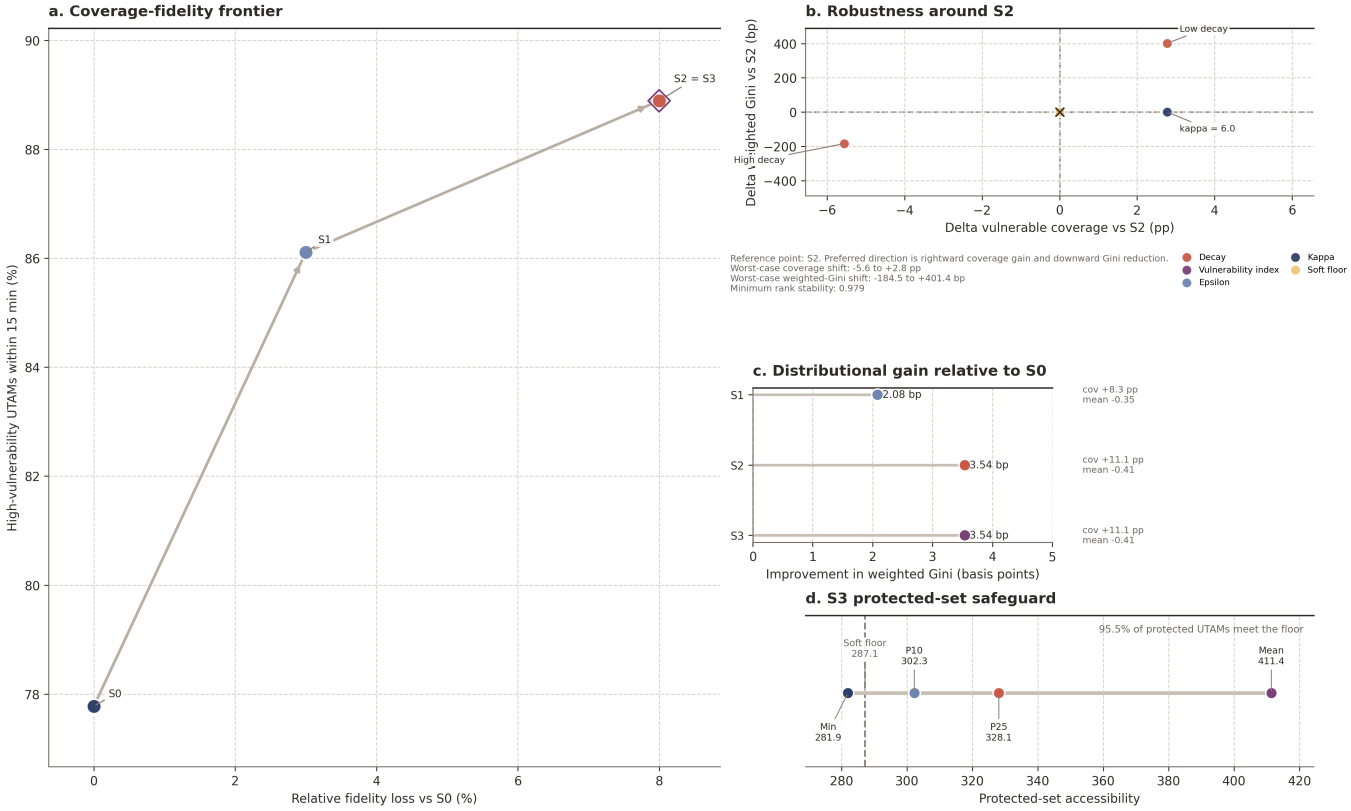


Figure 8: Policy trade-off synthesis across allocation scenarios. Panel (a) shows the coverage-fidelity frontier, where the move from S0 to S1 delivers the largest marginal gain in vulnerable-zone coverage and the move from S1 to S2/S3 yields a smaller additional improvement at higher fidelity cost. Panel (b) summarizes robustness around S2 across alternative decay, vulnerability-index, epsilon, kappa and soft-floor specifications. Panel (c) reports weighted-Gini improvements relative to S0 alongside gains in vulnerable coverage and mean accessibility for high-vulnerability areas. Panel (d) reports the lower-tail safeguard for the protected set under S3. S2 and S3 overlap on the frontier because the soft-floor safeguard is non-binding in the reproduced allocation.

zones, reduces the vulnerability-weighted accessibility Gini from 0.197041 to 0.196896, and raises coverage of vulnerable UTAMs within 15 minutes from 86.11% to 88.89%, at an 8.0% fidelity loss relative to the priority-matching baseline.

The soft-floor extension is informative as a robustness check but does not change the reproduced allocation relative to S2: S3 coincides with S2, and 95.45% of protected zones already satisfy the calibrated minimum-service floor. For Bogotá and similar data-constrained cities, the practical implication is staged deployment: begin with a demand-preserving baseline, relax toward territorial equity, add vulnerability weighting when the proxy evidence is credible, and use soft floors first as monitoring safeguards in the reproduced configuration, tightening them into stricter service guarantees only after additional sensitivity analysis.

Supplementary material

Notation tables and supplementary figures for graph learning, target extraction, optimization, equity, and robustness checks are provided in the separate supplementary appendix submitted with the manuscript.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used generative AI and AI-assisted technologies in order to improve the readability and language of the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

Code, figure-generation scripts, and the shareable processed data and derived non-sensitive outputs used to reproduce the analysis are archived in a blinded replication package prepared for double-anonymized review. Repository identifiers are omitted from the manuscript and can be supplied directly to the editor. The full raw input datasets cannot be deposited publicly because some sources are subject to licensing, institutional-access, or privacy-related restrictions.

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